COLLEGE CODE:9602

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CASE STUDY 2: MediCare Hospital – Patient & Operations Analytics

SUBMITTED BY,

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Case Study 2: MediCare Hospital - Patient & Operations Analytics

Industry: Healthcare

Modules:

- Reporting
- Exploration
- Dashboard
- Story Making
- Website

Real-Time Business Scenario:

MediCare Hospital is a multi-speciality hospital with 500 beds, 40 departments, and over 1,200 staff members. The hospital administration is concerned about rising patient wait times, bed occupancy rates exceeding 90%, and increasing re-admission rates. The Chief Medical Officer wants a data driven view of hospital operations, doctor performance, and patient outcomes to improve service quality and resource allocation using IBM Cognos Analytics.

Project Description:

This project focuses on developing a data-driven healthcare analytics dashboard to monitor and improve hospital operations. Using IBM Cognos Analytics, reports, visualizations, and dashboards were created to analyze key metrics such as bed occupancy rate, patient wait time, readmission rate, and patient satisfaction. The system highlights ward-level occupancy trends over the last six months and uses conditional formatting to identify high-risk areas, supporting better resource allocation and decision-making.

Exploratory analysis was conducted to understand the relationship between patient satisfaction, wait time, and length of stay, leading to actionable recommendations for improving efficiency and care quality. Additionally, a web-based dashboard was developed using AI-assisted tools, enabling interactive data visualization and real-time monitoring. This project demonstrates the effective use of analytics and visualization in supporting data-driven healthcare management.

1. List Report

Create a List Report showing Patient ID, Department, Doctor Name, Length of Stay, and Bill Amount. Apply a Header with the hospital name and report generation date, Footer with confidentiality notice and page number. Use Calibri font size 11, Dark Red color for re-admitted patients using Conditional Styling. Add a prompt to filter by Department.

IBM Cognos Analytics | 1. Conditional Styles

HTML | Mode: Interactive View

Zenith Health Center- Patient Admission and Billing Report - 5/4/2026

Patient ID	Department	Doctor Name	Length of Stay (days)	Re-admission Flag	Bill Amount
PT0001	Orthopaedics	Dr. Reddy	7	No	144893
PT0003	ENT	Dr. Rao	9	No	184528
PT0006	General Medicine	Dr. Mehta	7	Yes	51993
PT0010	Orthopaedics	Dr. Iyer	14	No	107015
PT0011	Dermatology	Dr. Iyer	10	No	90557
PT0012	Dermatology	Dr. Singh	13	Yes	82431
PT0014	Paediatrics	Dr. Rao	10	Yes	90047
PT0015	ENT	Dr. Mehta	8	No	21585
PT0017	Oncology	Dr. Nair	3	No	89060
PT0024	Paediatrics	Dr. Reddy	8	No	111180
PT0025	Oncology	Dr. Patel	5	Yes	189486
PT0030	Orthopaedics	Dr. Mehta	5	No	145930
PT0032	ENT	Dr. Mehta	12	No	191319
PT0035	Dermatology	Dr. Nair	1	No	197079
PT0036	Cardiology	Dr. Patel	9	No	39234
PT0040	ENT	Dr. Nair	3	No	39903
PT0043	Dermatology	Dr. Singh	11	Yes	23615

Top | Page up | Page down | Bottom

IBM Cognos Analytics | 1. Conditional Styles

HTML | Mode: Interactive View

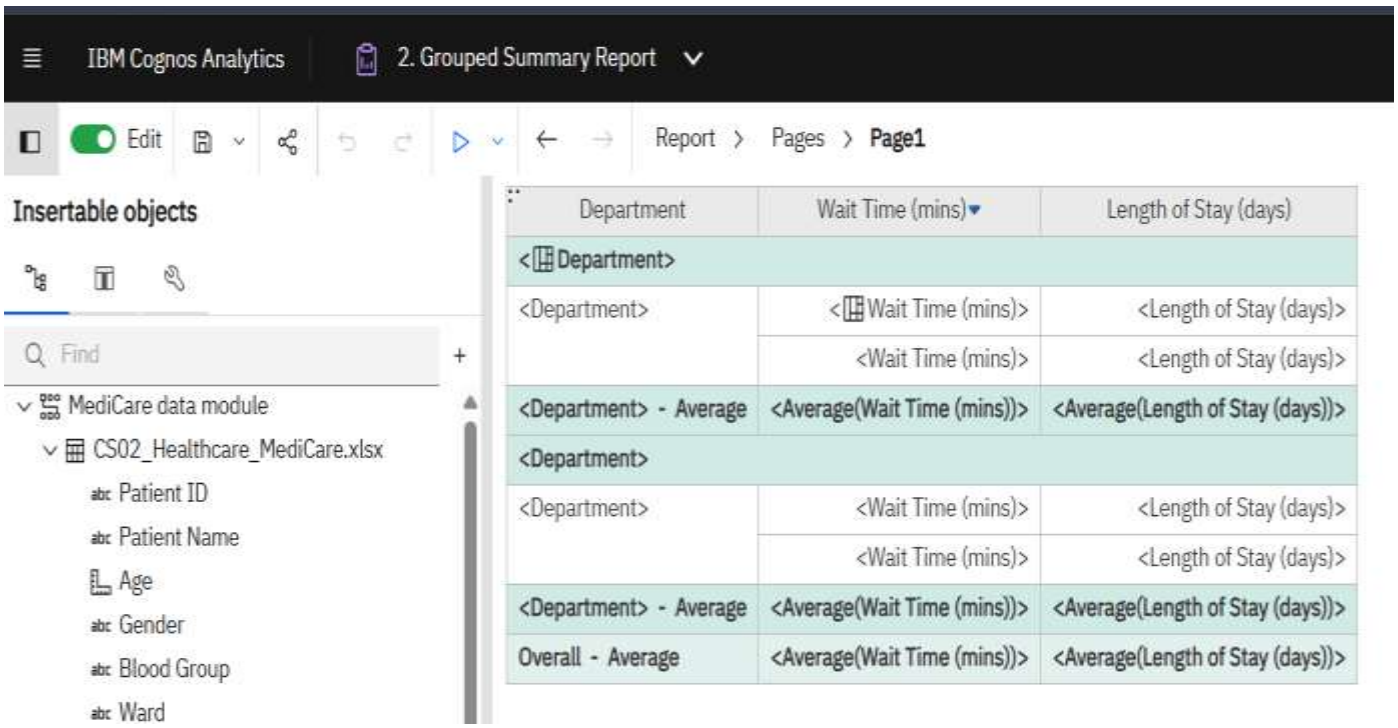
Zenith Health Center- Patient Admission and Billing Report - 5/4/2026

Patient ID	Department	Doctor Name	Length of Stay (days)	Re-admission Flag	Bill Amount
PT0157	Dermatology	Dr. Singh	7	Yes	140398
PT0158	Paediatrics	Dr. Mehta	6	No	155395
PT0160	Dermatology	Dr. Reddy	14	No	133661
PT0162	Dermatology	Dr. Singh	5	No	75623
PT0165	Dermatology	Dr. Patel	2	No	92728
PT0166	Dermatology	Dr. Mehta	10	No	175358
PT0179	Dermatology	Dr. Mehta	7	Yes	121119
PT0180	Paediatrics	Dr. Rao	2	Yes	68724
PT0181	ENT	Dr. Reddy	1	Yes	98196
PT0187	Orthopaedics	Dr. Nair	9	Yes	114039
PT0192	Dermatology	Dr. Nair	3	No	101228
PT0195	Gynaecology	Dr. Iyer	12	No	45159
PT0004	Paediatrics	Dr. Patel	11	Yes	142348
PT0007	Dermatology	Dr. Iyer	7	No	153111
PT0008	Oncology	Dr. Mehta	2	No	118746
PT0009	Neurology	Dr. Iyer	15	Yes	152720
PT0013	General Medicine	Dr. Iyer	8	No	75717
PT0018	Oncology	Dr. Mehta	6	No	152810
PT0020	Cardiology	Dr. Iyer	7	Yes	14637

Top | Page up | Page down | Bottom

2. Grouped Summary Report

Build a Grouped Summary Report showing Average Wait Time and Average Length of Stay by Department. Sort departments by Average Wait Time in descending order. Add subtotals per department group. Use the Report Summarize feature to show overall hospital averages at the report footer.



IBM Cognos Analytics | 2. Grouped Summary Report

HTML

Department	Wait Time (mins)	Length of Stay (days)
Cardiology		
Cardiology	6	219
Cardiology - Average	**6**	**219**
Dermatology		
Dermatology	19	180
Dermatology - Average	**19**	**180**
ENT		
ENT	8	148
ENT - Average	**8**	**148**
General Medicine		
General Medicine	9	189
General Medicine - Average	**9**	**189**
Gynaecology		
Gynaecology	6	155
Gynaecology - Average	**6**	**155**
Neurology		
Neurology	6	117
Neurology - Average	**6**	**117**
Oncology		
Oncology	11	167
Oncology - Average	**11**	**167**

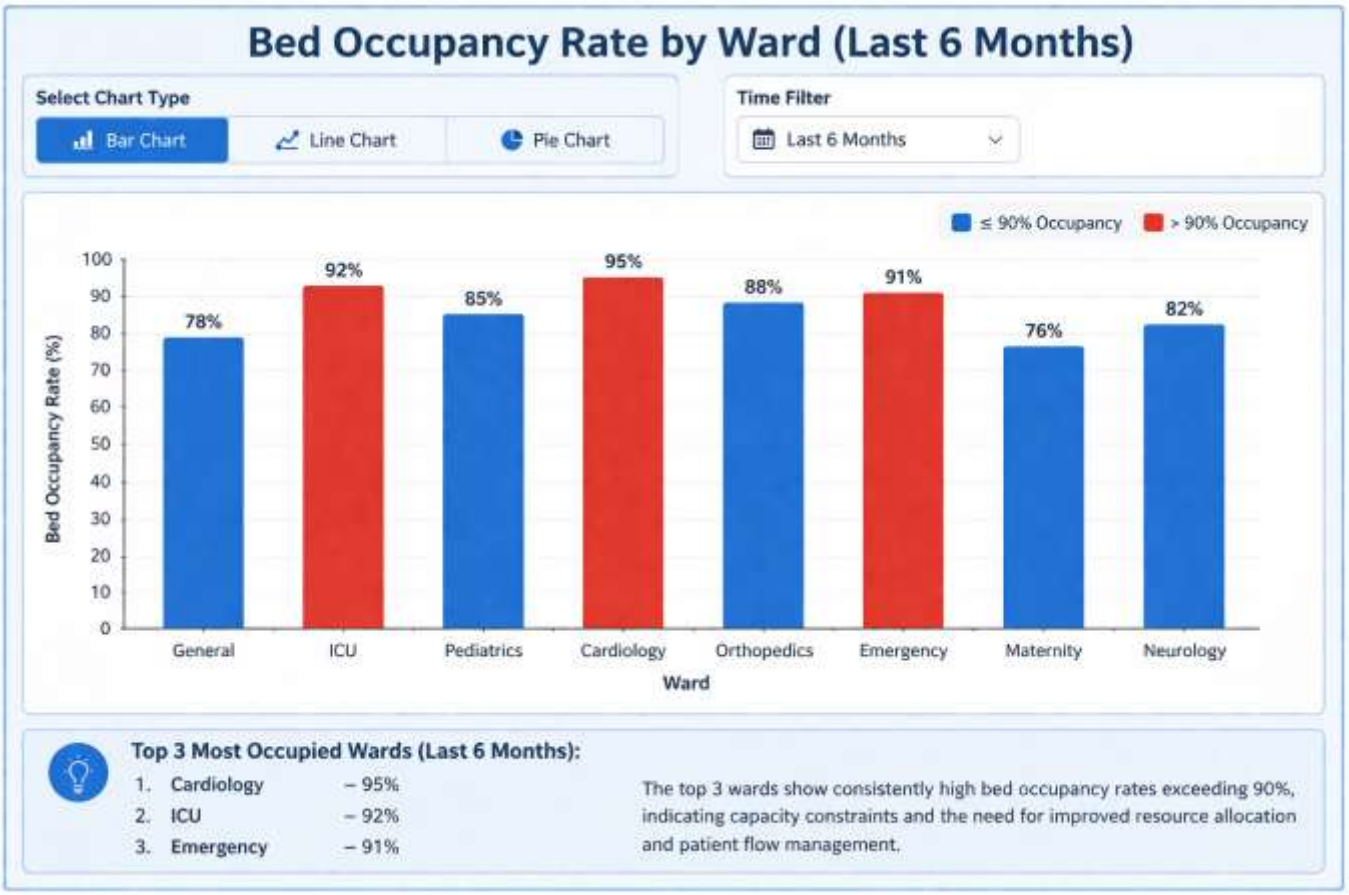
IBM Cognos Analytics | 2. Grouped Summary Report

HTML

ENT		
ENT	8	148
ENT - Average	**8**	**148**
General Medicine		
General Medicine	9	189
General Medicine - Average	**9**	**189**
Gynaecology		
Gynaecology	6	155
Gynaecology - Average	**6**	**155**
Neurology		
Neurology	6	117
Neurology - Average	**6**	**117**
Oncology		
Oncology	11	167
Oncology - Average	**11**	**167**
Orthopaedics		
Orthopaedics	6	227
Orthopaedics - Average	**6**	**227**
Paediatrics		
Paediatrics	6	143
Paediatrics - Average	**6**	**143**
Overall - Average	**8.55555556**	**171.66666667**

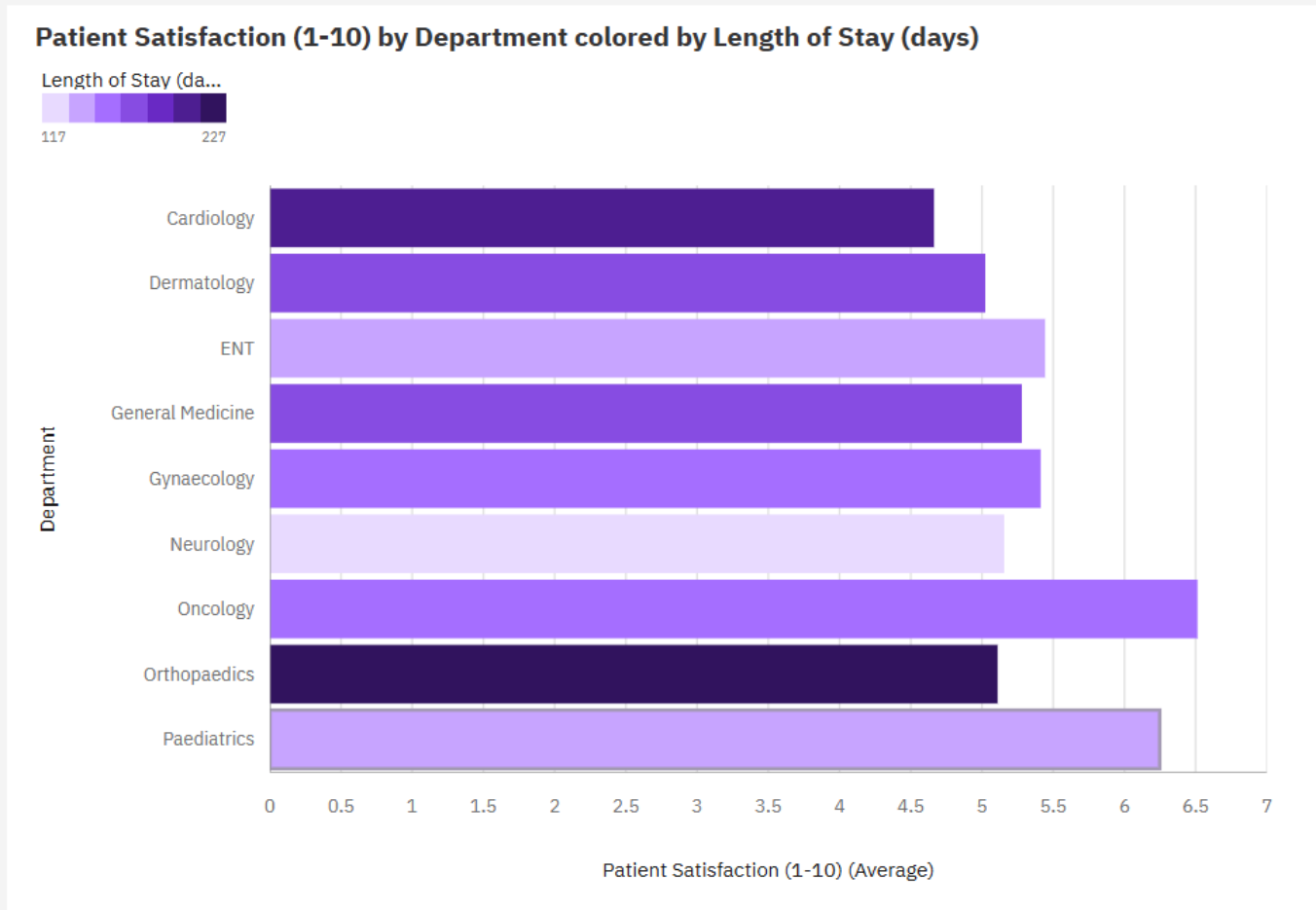
3. Visualization Report

Create a Visualization Report using Report Visualize Prompt where users can select a chart type to view Bed Occupancy Rate by Ward over the last 6 months. Apply Conditional Styling to highlight any ward exceeding 90% occupancy in Red. Add a text summary of the top 3 most occupied wards.

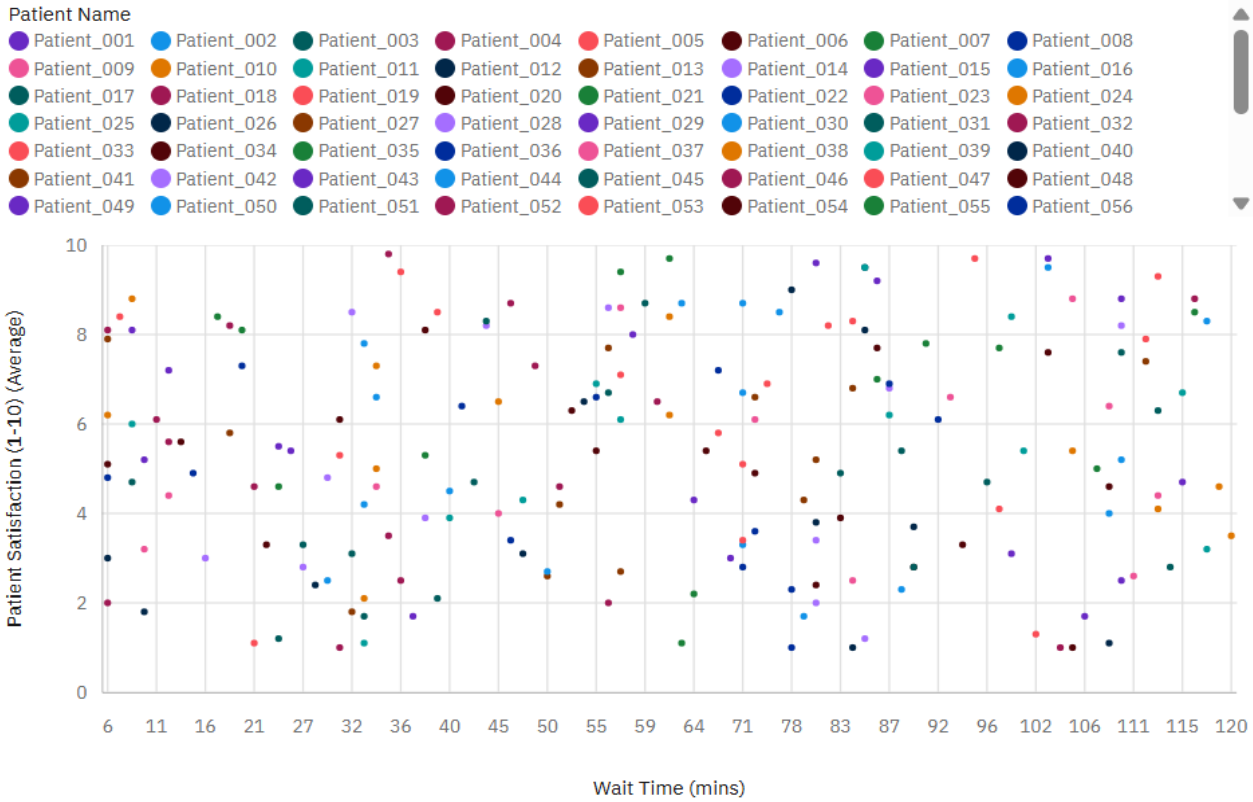


4. Exploration

Use Cognos Exploration to analyze the Patient Satisfaction Score dataset. Explore correlations between Wait Time and Patient Satisfaction Score. Identify which departments have the lowest satisfaction scores and what common factors (doctor, wait time, length of stay) contribute to them. Document your exploration findings.



Patient Satisfaction (1-10) by Wait Time (mins) colored by Patient Name



Department, Doctor Name, Length of Stay (days), Wait Time (mins) and Patient Satisfaction (1-10)

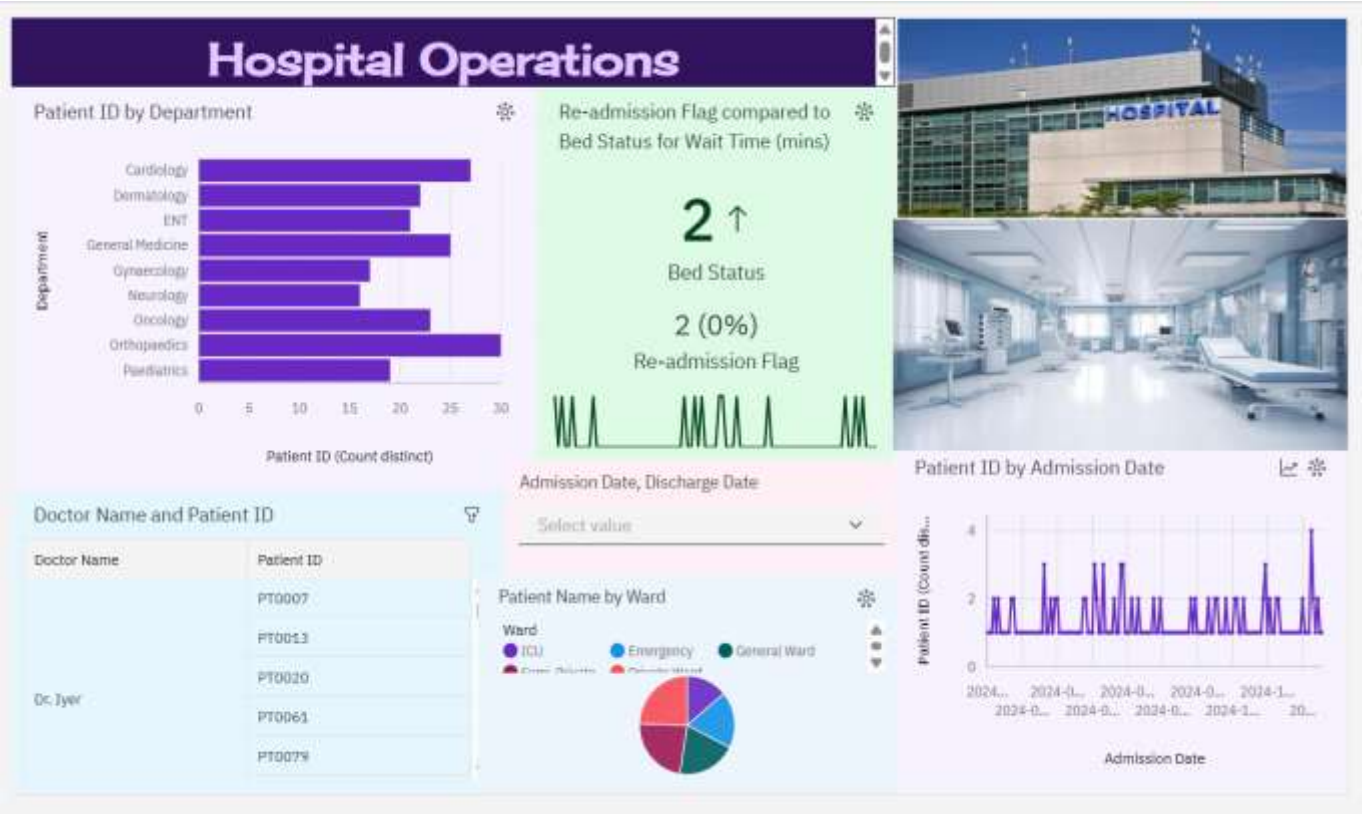
Department	Doctor Name	Length of Stay (days)	Wait Time (mins)	Patient Satisfaction (1...
Cardiology	Dr. Iyer	69	12	5.563
	Dr. Mehta	17	16	3.1
	Dr. Nair	6	81	2.4
	Dr. Patel	20	56	1.5
	Dr. Rao	36	39	3.8
	Dr. Reddy	11	24	5.1
	Dr. Sharma	31	6	7.333
	Dr. Singh	29	27	5.4
Summary		**219**	**6**	**4.663**
Dermatology	Dr. Iyer	36	19	3.58
	Dr. Mehta	17	78	4.95
	Dr. Nair	16	28	6.333
	Dr. Patel	14	48	7
	Dr. Reddy	32	40	6.2
	Dr. Singh	65	35	4.443

Department, Doctor Name, Length of Stay (days), Wait Time (mins) and Patient Satisfaction (1-10)

Department	Doctor Name	Length of Stay (days)	Wait Time (mins)	Patient Satisfaction (1...
Orthopaedics	Dr. Rao	50	20	5.633
	Dr. Reddy	27	32	4.4
	Dr. Sharma	28	23	3.625
	Dr. Singh	17	27	4.95
Summary		**227**	**6**	**5.11**
Paediatrics	Dr. Iyer	20	7	8.25
	Dr. Mehta	23	32	5.7
	Dr. Nair	13	6	7.15
	Dr. Patel	18	12	4.15
	Dr. Rao	40	21	5.74
	Dr. Reddy	18	34	7.05
	Dr. Sharma	9	110	8.2
	Dr. Singh	2	6	5.75
Summary		**143**	**6**	**6.247**
Summary		**1545**	**6**	**5.395**

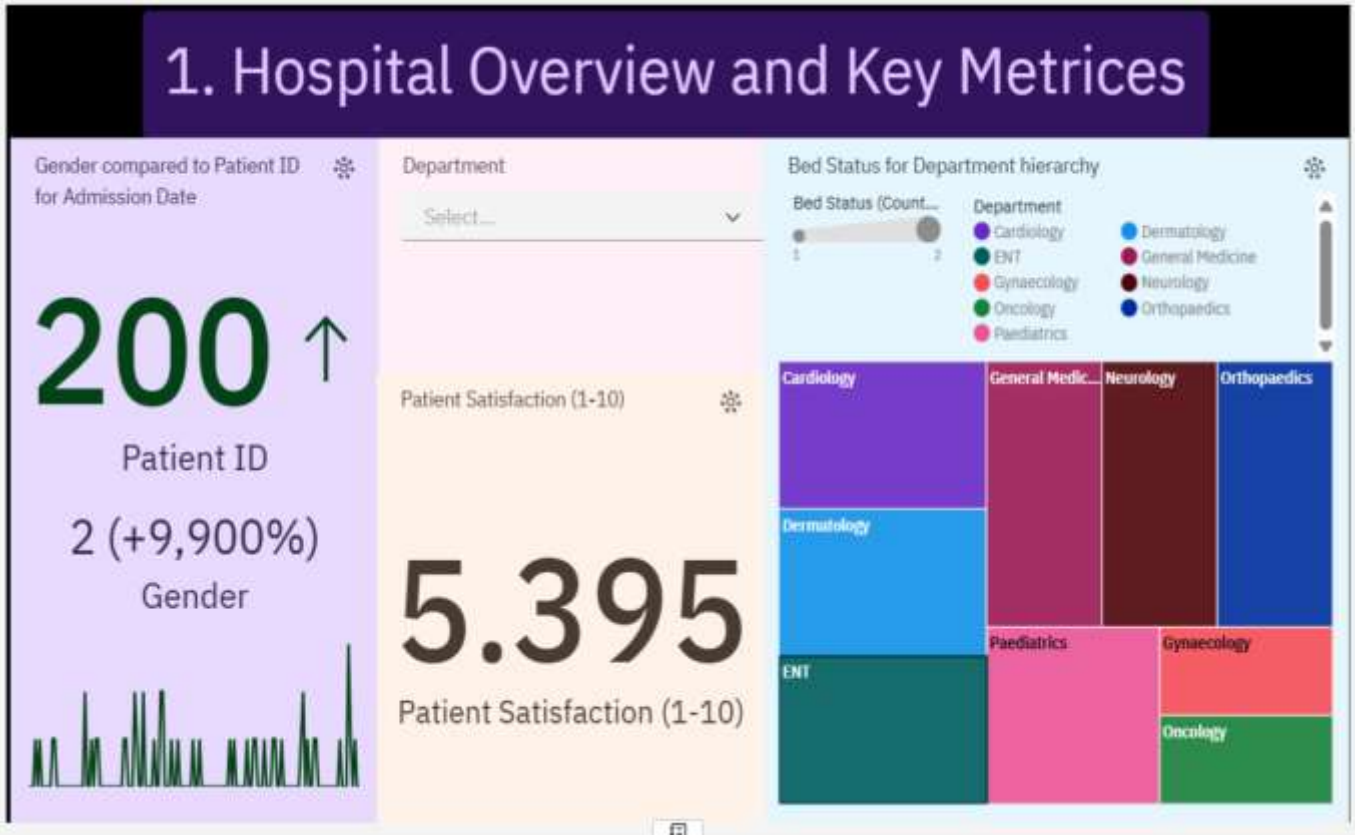
5. Hospital Operations Dashboard

Build a Hospital Operations Dashboard with: (1) KPI tiles for Bed Occupancy Rate, Average Wait Time, and Re-admission Rate, (2) A Bar chart for Patient Count by Department, (3) A Line chart for Monthly Admissions Trend, (4) A Table of Top 5 Doctors by Patient Count. Include a Date Range filter for the entire dashboard.



6. Hospital Story

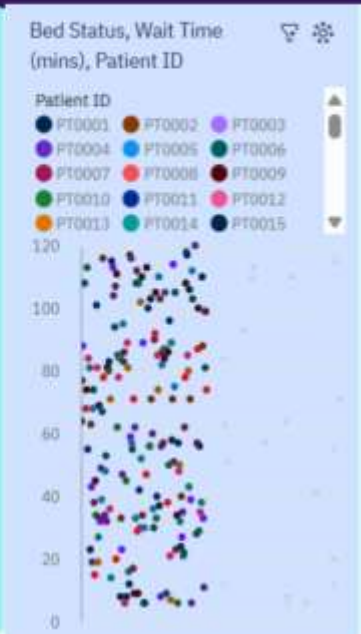
Create a Cognos Story for the Hospital Board Meeting with 5 scenes: Scene 1 — Hospital Overview & Key Metrics, Scene 2 — Department-wise Patient Load, Scene 3 — Wait Time & Bed Occupancy Crisis Areas, Scene 4 — Patient Satisfaction Analysis, Scene 5 — Action Plan & Recommendations. Export the story as a PDF for board presentation.


1. Hospital Overview and Key Metrices
Gender compared to Patient ID for Admission Date
200 ↑
Patient ID
2 (+9,900%)
Gender
Department
Select...
Patient Satisfaction (1-10)
5.395
Patient Satisfaction (1-10)
Bed Status for Department hierarchy
Bed Status (Count...
Department
Cardiology
Dermatology
ENT
General Medicine
Gynaecology
Neurology
Oncology
Orthopaedics
Paediatrics
Cardiology
General Medic...
Neurology
Orthopaedics
Dermatology
Paediatrics
Gynaecology
ENT
Oncology

2. Department wise Patient Load
"This visualization shows the distribution of patient load across different hospital departments. It helps identify which departments experience higher demand, enabling better resource allocation and staffing decisions."
Patient ID compared to Bed Status by Department
Patient ID (Count distinct)
Bed Status (Count distinct)
Cardiology
Dermatology
ENT
General Medicine
Gynaecology
Neurology
Oncology
Orthopaedics
Paediatrics
Department
0
10
20
30
Patient ID (Count distinct)
Column 1
Patient ID (Count distinct)
Column 2
Department (Count distinct)
200
150
100
50
0
10
8
6
4
2
0
Patient ID (Count distinct)
Department (Count distinct)

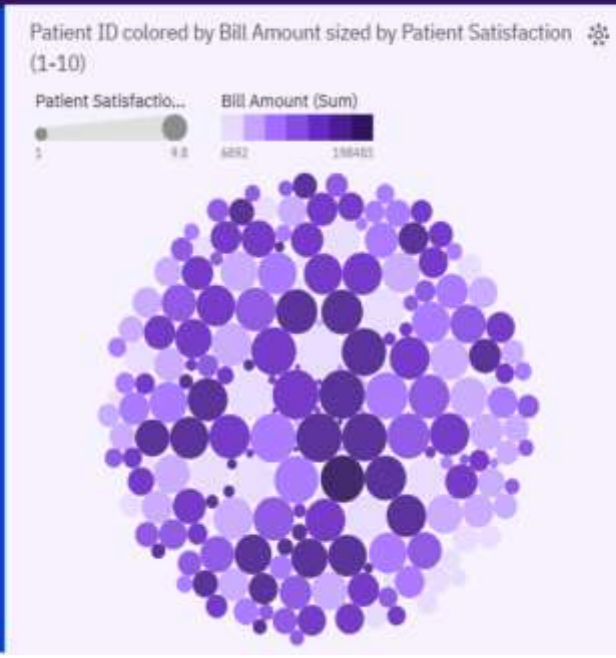
3. Wait Time and Bed Occupancy Crisis Areas

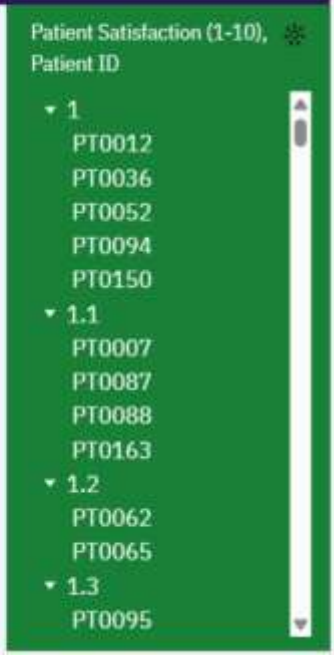
"Wards with occupancy levels above 90% show significantly higher patient wait times, suggesting operational inefficiencies and capacity strain."



4. Patient Satisfaction Analysis

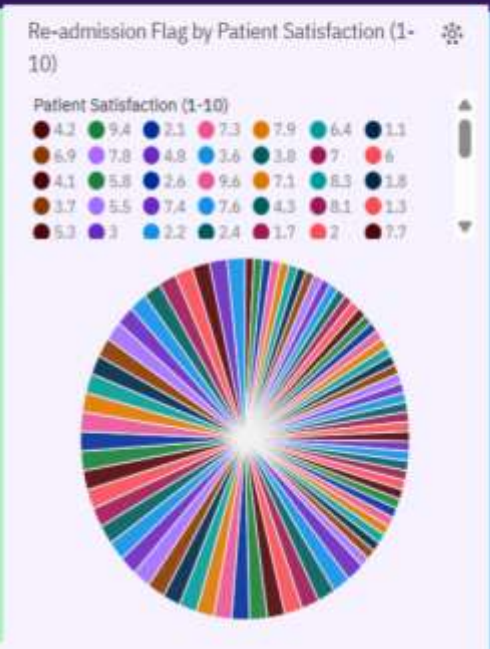
"Improving operational efficiency, especially in high-load departments, can significantly enhance overall patient satisfaction."





5. Action Plan & Recommendations

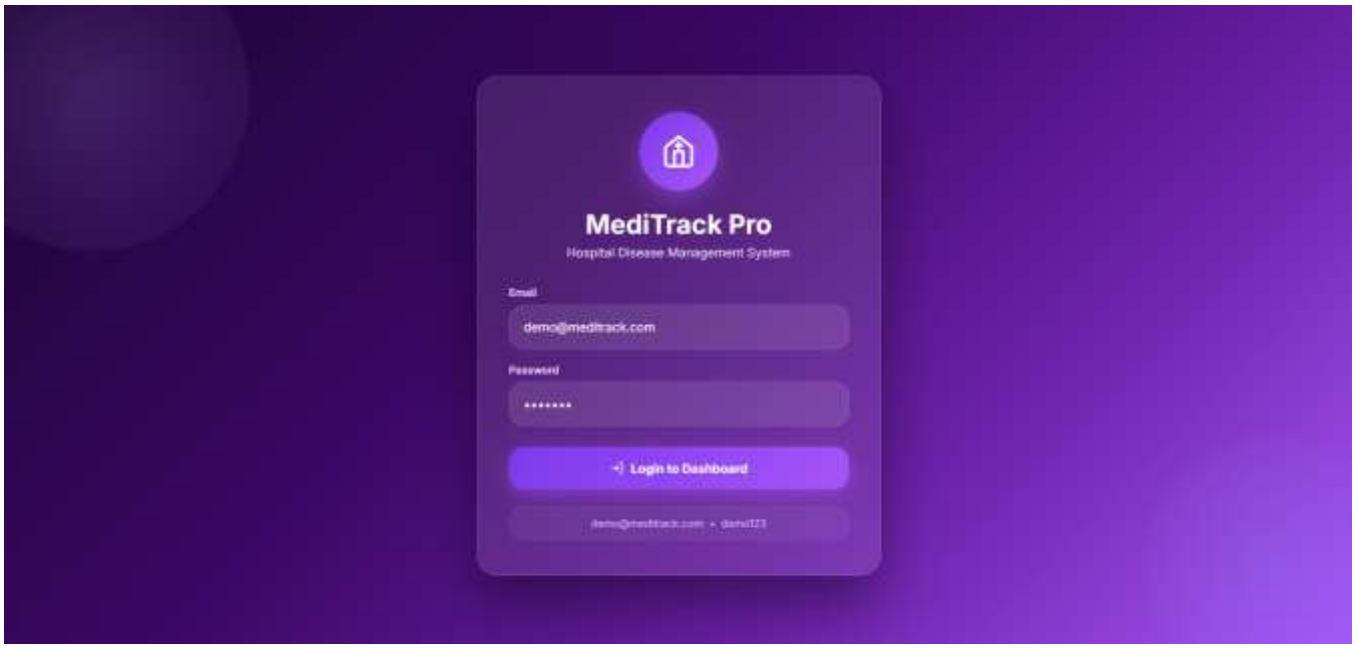
"A data-driven approach focusing on capacity management, process optimization, and patient-centric care can significantly improve operational efficiency and patient satisfaction."

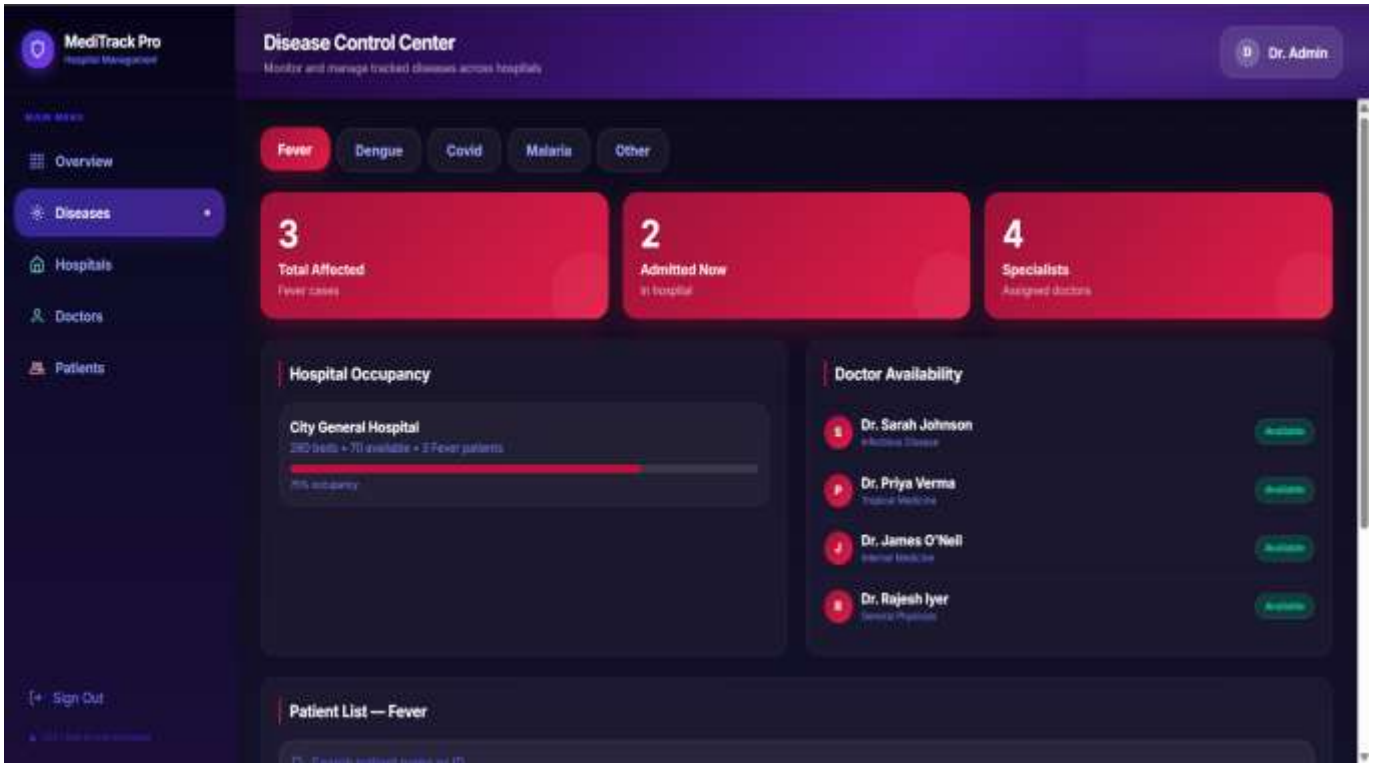


WEBSITE: https://medi-track-system--shamijabalin.replit.app/

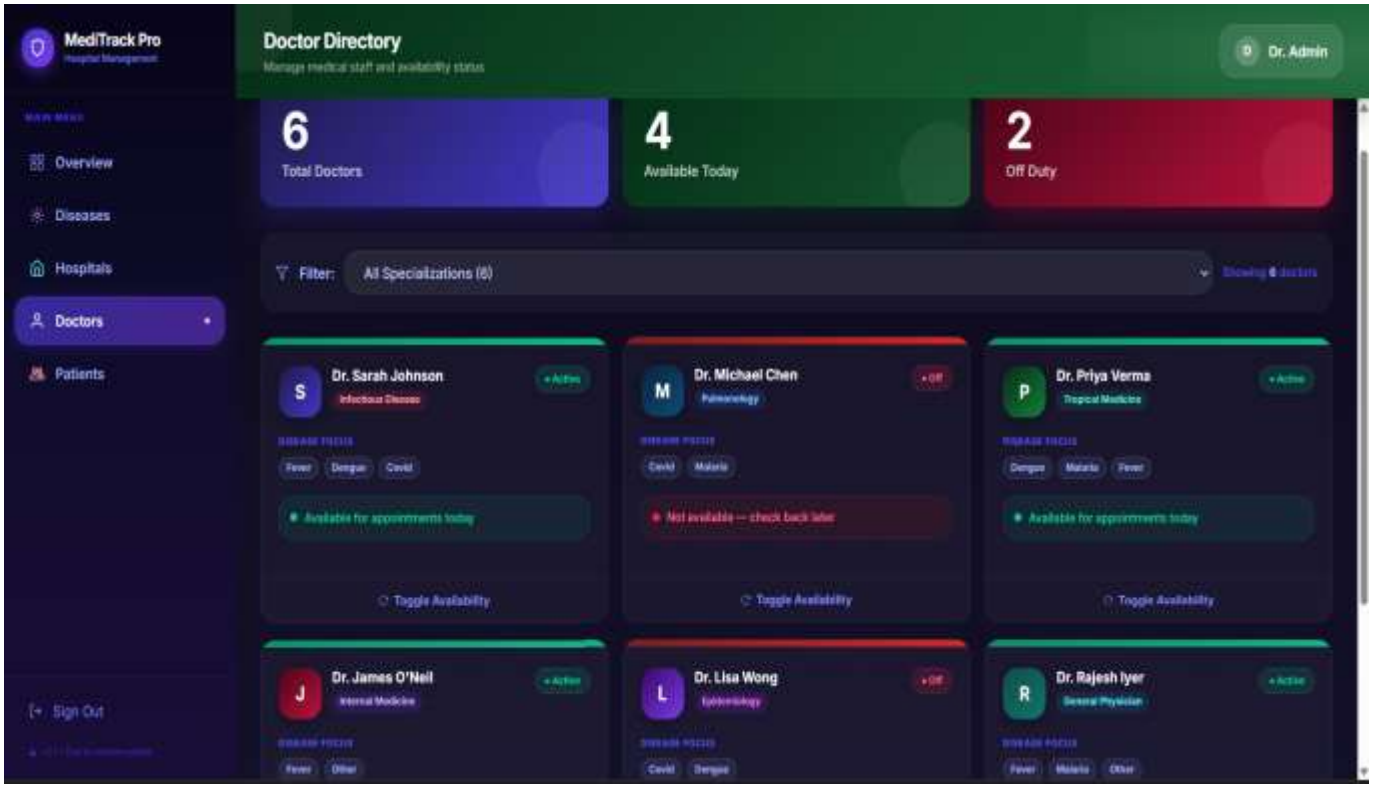
MEDITRACK PRO
Smart Hospital Management System

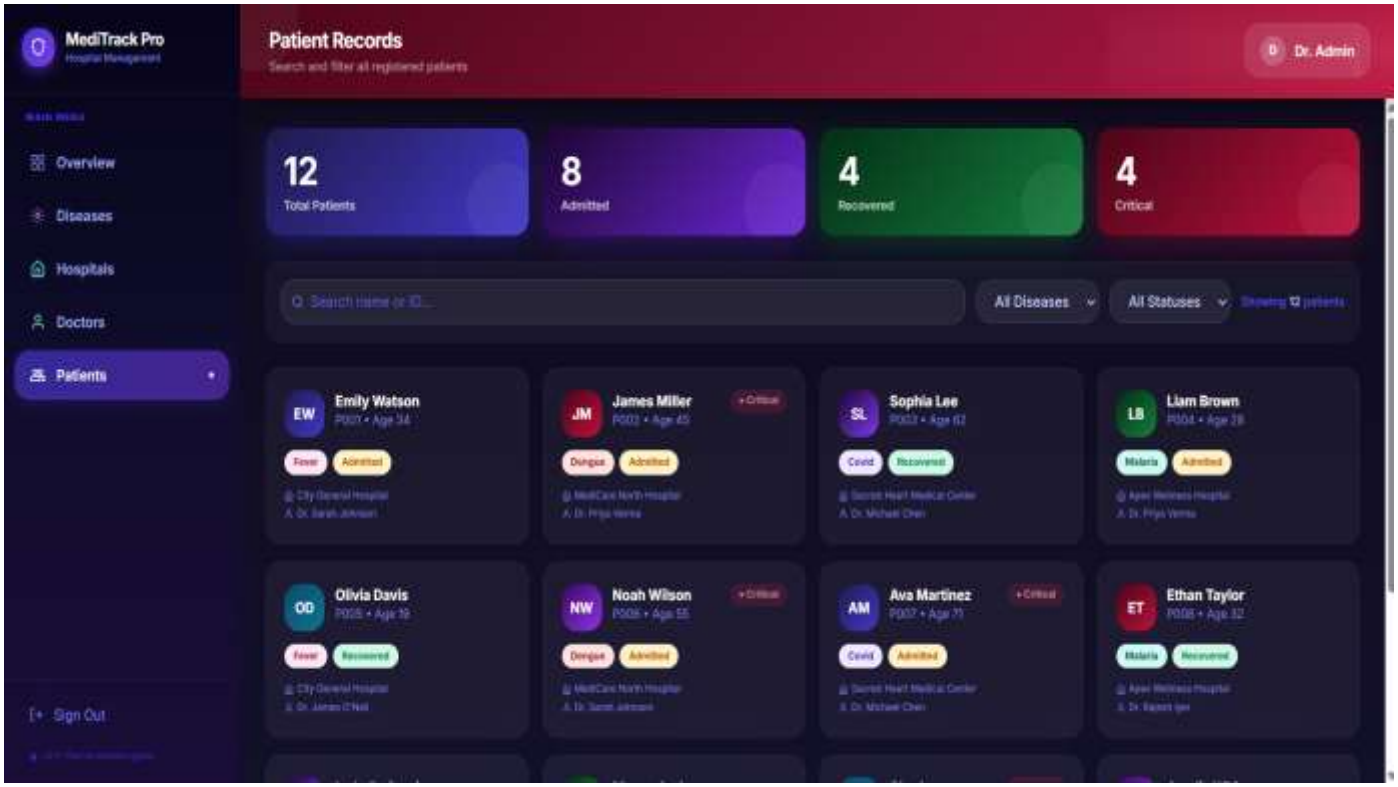
A data-driven platform to monitor hospital performance, patient care, and operational efficiency through interactive dashboards.




MediTrack Pro
Disease Control Center
Dr. Admin
Overview
Diseases
Hospitals
Doctors
Patients
Fever
Dengue
Covid
Malaria
Other
3
Total Affected
2
Admitted Now
4
Specialists
Hospital Occupancy
City General Hospital
Doctor Availability
Dr. Sarah Johnson
Dr. Priya Verma
Dr. James O'Neil
Dr. Rajesh Iyer
Sign Out
Patient List — Fever

MediTrack Pro
Hospital Management
Dr. Admin
Overview
Diseases
Hospitals
Doctors
Patients
City General Hospital
Hospital #1
280
210
70
Occupancy Rate
75%
Total registered patients: 134
MediCare North Hospital
Hospital #2
420
305
115
Occupancy Rate
73%
Total registered patients: 198
Sacred Heart Medical Center
Hospital #3
350
278
72
Occupancy Rate
79%
Apex Wellness Hospital
Hospital #4
200
142
58
Occupancy Rate
71%
Sign Out





GITHUB Link: https://github.com/jabalinshami2005-netizen/IBM-Cognos-Analytics